CARDIOTWIN
SMART HEALTH MONITORING

CardioTwin: Smart Wearable Health Monitoring

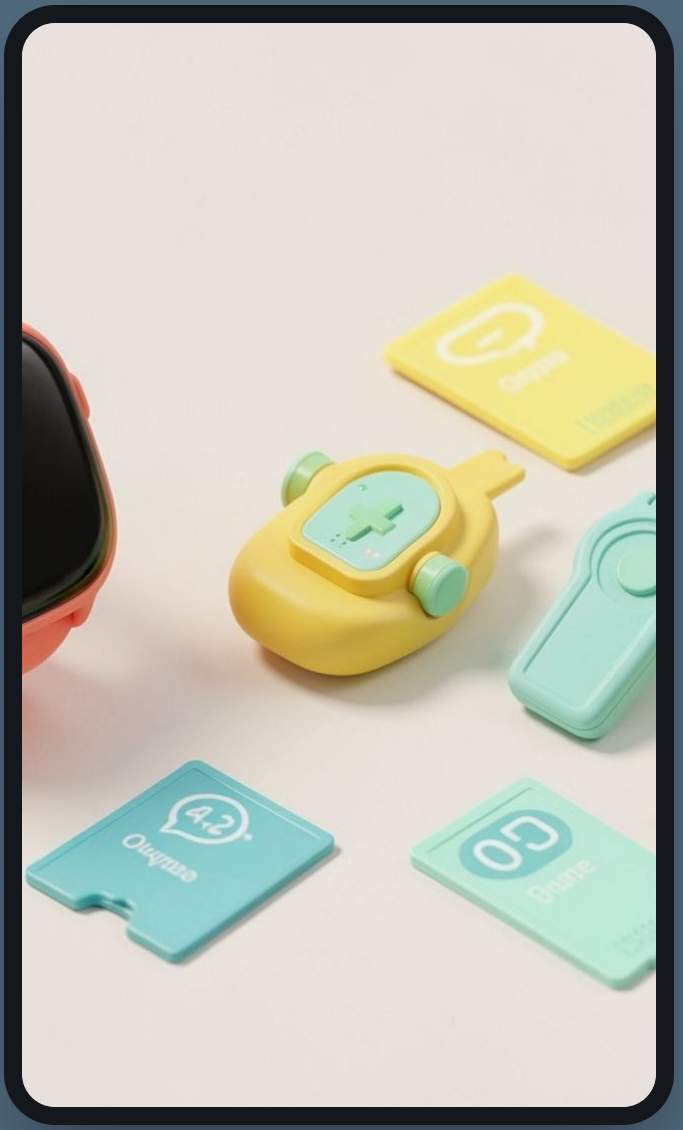
ANAGHA S KUMAR

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THE PROBLEM

Wearable data is everywhere. Meaning is harder to find.

- Health data is scattered across wearable applications.
- Continuous tracking does not always provide meaningful insights.
- Identifying changes in heart rate and oxygen saturation manually is difficult.
- Existing advanced monitoring solutions may be expensive.



Signals without a shared view.

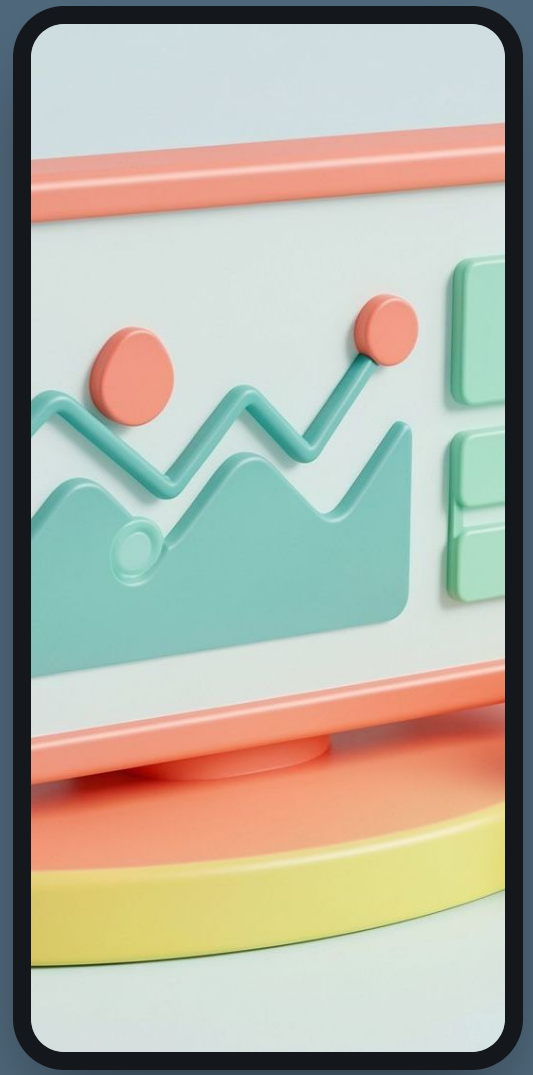
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THE WORKFLOW

How CardioTwin Works

Data Collection	Obtain heart rate, SpO_2, steps, sleep, and activity data.
Data Processing	Clean, validate, and organize readings using Python.
Pattern Analysis	Calculate summaries and identify selected changes in readings.
Visualization	Display trends, activity comparisons, and daily summaries.
Alerts	Flag readings that cross predefined thresholds for review.

Current Status: The prototype uses synthetic data to test the analysis and visualization workflow. Real wearable integration is planned.



A modular path from readings to review.

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THE PIPELINE

CardioTwin System Architecture

Layer	System components
Data Source Layer	Wearable Device / Synthetic CSV Data
Data Integration Layer	Google Fit / Health Connect (planned integration)
Data Processing Layer	Python + Pandas Data Cleaning and Validation
Analysis Layer	Trend Analysis + Threshold-Based Flags
Visualization Layer	Streamlit + Plotly Dashboard
Output	Health Trends, Daily Summaries, and Readings for Review

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THE VALUE

Affordable and Scalable Monitoring



Low-cost starting point

Uses existing wearable devices instead of specialized monitoring hardware. Uses CSV data for initial testing, reducing hardware and setup costs.



Open-source tools

Built with open-source tools: Python, Pandas, Streamlit, and Plotly.



Scale over time

Modular architecture allows future integration with different wearable platforms. Can be extended with cloud storage and automated synchronization.

Value proposition: A software-first approach that reduces additional hardware requirements while supporting future expansion.

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THE NEXT LAYER

AI-Powered Cardiovascular Insights

PLANNED AI CAPABILITIES



Learn a baseline

Learn personalized baselines from historical readings.



Spot anomalies

Detect anomalies and unusual patterns.



Explain change

Explain personalized changes from an individual's baseline.



Read trends

Analyze heart rate, SpO_2, sleep, and activity trends.

Future Development

Real wearable data integration · Automated data synchronization · AI-based pattern recognition and risk assessment · Secure cloud-based monitoring and data storage.

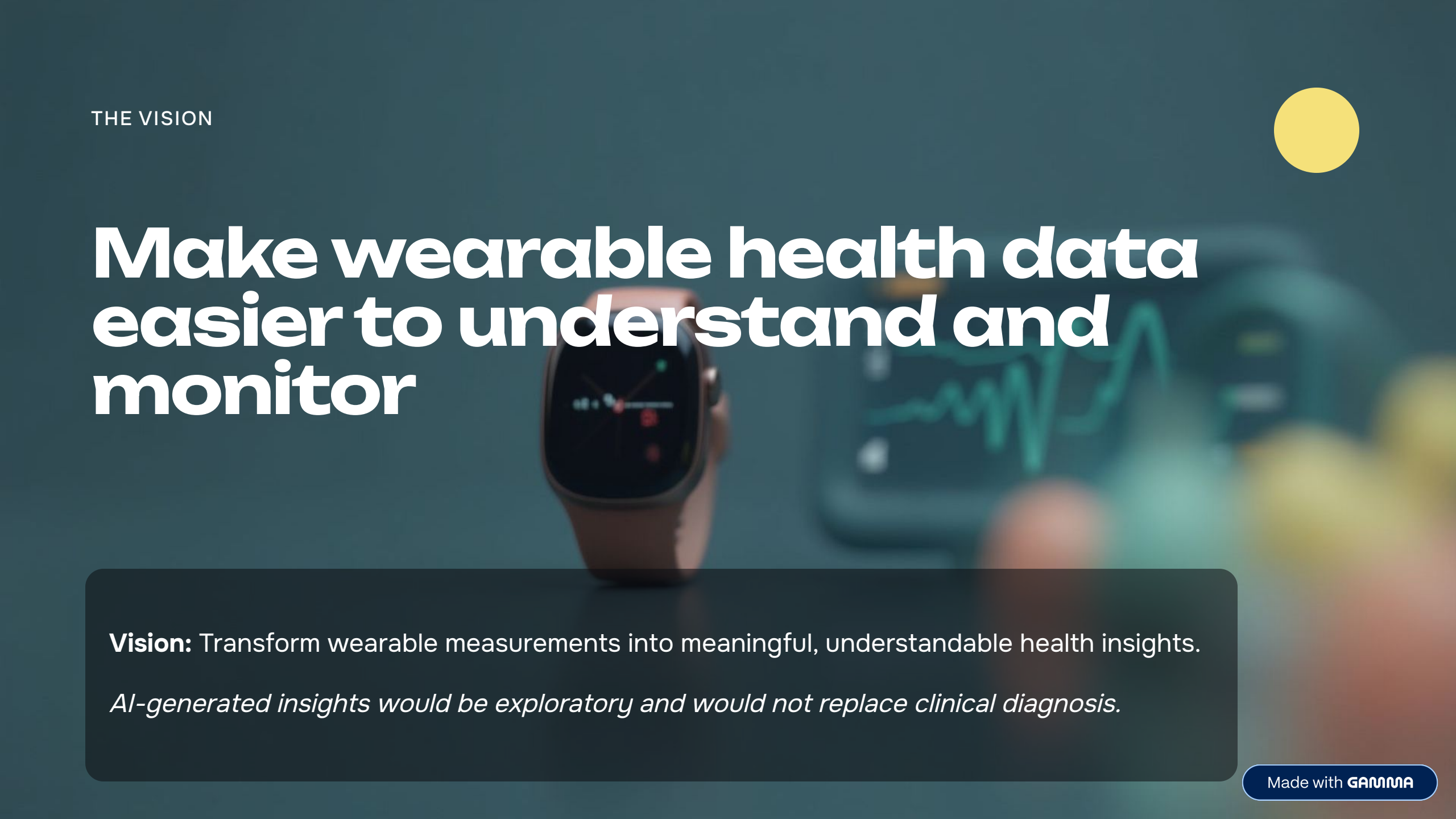
Prototype boundary

Testing uses synthetic data. AI-generated insights support review, not clinical diagnosis.

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THE VISION

Make wearable health data easier to understand and monitor



Vision: Transform wearable measurements into meaningful, understandable health insights.

AI-generated insights would be exploratory and would not replace clinical diagnosis.

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